

5.4.9

EE25BTECH11014 - Bhoomika Lokesh

Question: Using elementary transformations, find the inverse of the following matrix

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Solution:

We know that,

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (1)$$

where $\mathbf{I}$ is a 2 X 2 identity matrix.

The following is the augmented matrix we get,

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right) \quad (2)$$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow \frac{R_2}{-2}} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array} \right) \quad (3)$$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array} \right) \quad (4)$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$$